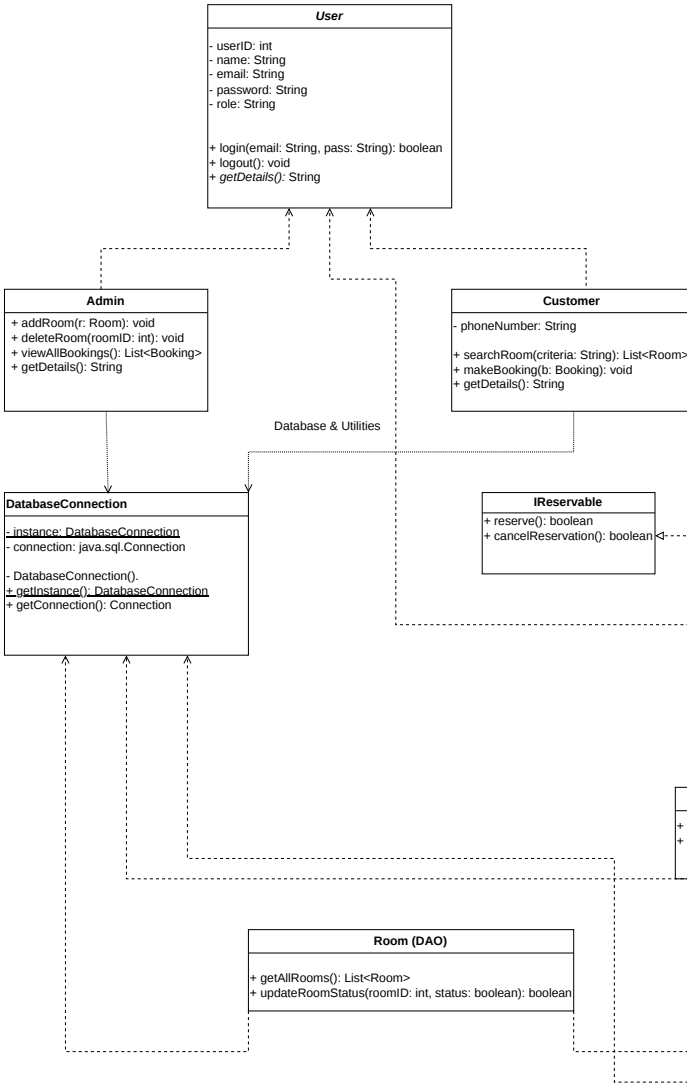
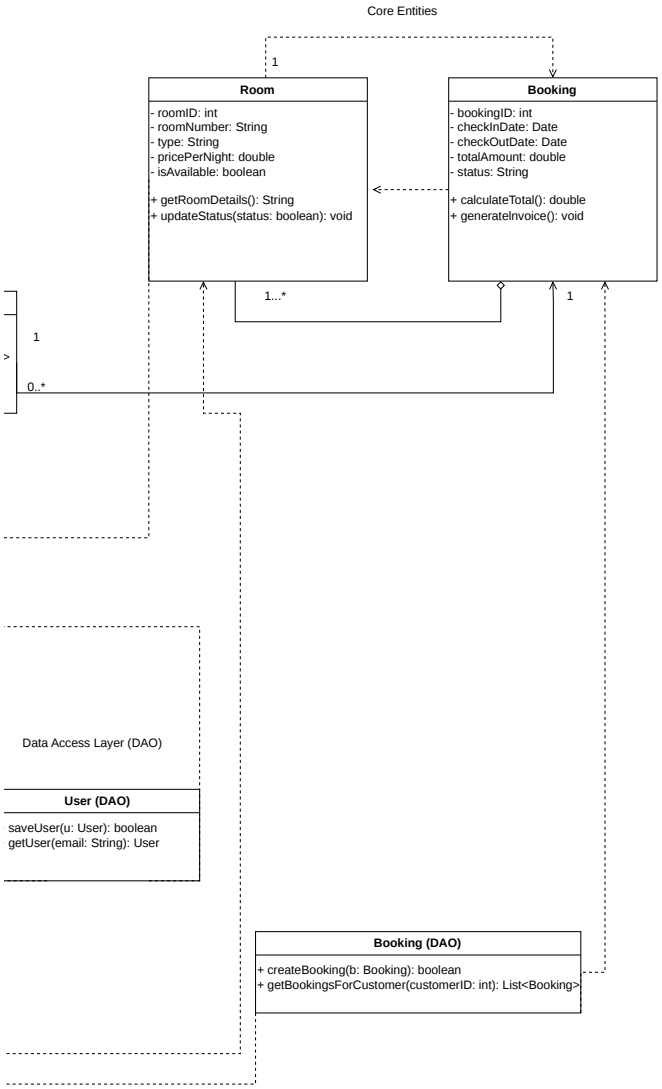

User
- userID: int
- name: String
- email: String
- password: String
- role: String
+ login(email: String, pass: String): boolean
+ logout(): void
+ getDetails(): String
Admin
+ addRoom(r: Room): void
+ deleteRoom(roomID: int): void
+ viewAllBookings(): List<Booking>
+ getDetails(): String
Customer
- phoneNumber: String
+ searchRoom(criteria: String): List<Room>
+ makeBooking(b: Booking): void
+ getDetails(): String
Database & Utilities
DatabaseConnection
- instance: DatabaseConnection
- connection: java.sql.Connection
- DatabaseConnection().
+ getInstance(): DatabaseConnection
+ getConnection(): Connection
IReservable
+ reserve(): boolean
+ cancelReservation(): boolean
Room (DAO)
+ getAllRooms(): List<Room>
+ updateRoomStatus(roomID: int, status: boolean): boolean


Core Entities
1
Room
- roomID: int
- roomNumber: String
- type: String
- pricePerNight: double
- isAvailable: boolean
+ getRoomDetails(): String
+ updateStatus(status: boolean): void
Booking
- bookingID: int
- checkInDate: Date
- checkOutDate: Date
- totalAmount: double
- status: String
+ calculateTotal(): double
+ generateInvoice(): void
1...*
1
1
0..*
Data Access Layer (DAO)
User (DAO)
saveUser(u: User): boolean
getUser(email: String): User
Booking (DAO)
+ createBooking(b: Booking): boolean
+ getBookingsForCustomer(customerID: int): List<Booking>